

- 8 The number of sides of a regular polygon B is triple the number of sides of regular polygon A .

The ratio of each interior angle of the polygon A to that of the polygon B is $3 : 4$.

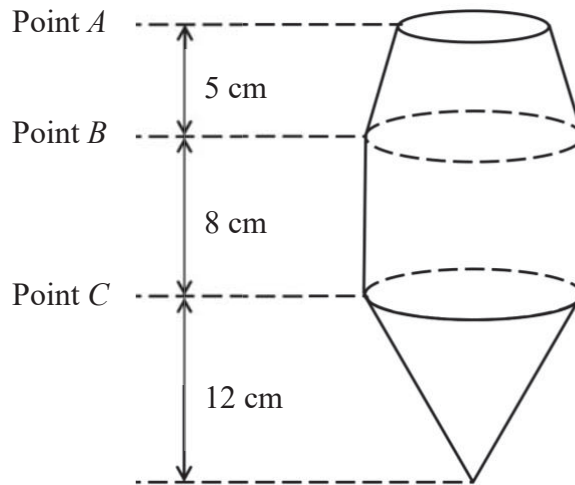
- (a) Find the number of sides of polygon A .

Answer [2]

- (b) Hence, find the size of each exterior angle of polygon B .

Answer $^{\circ}$ [2]

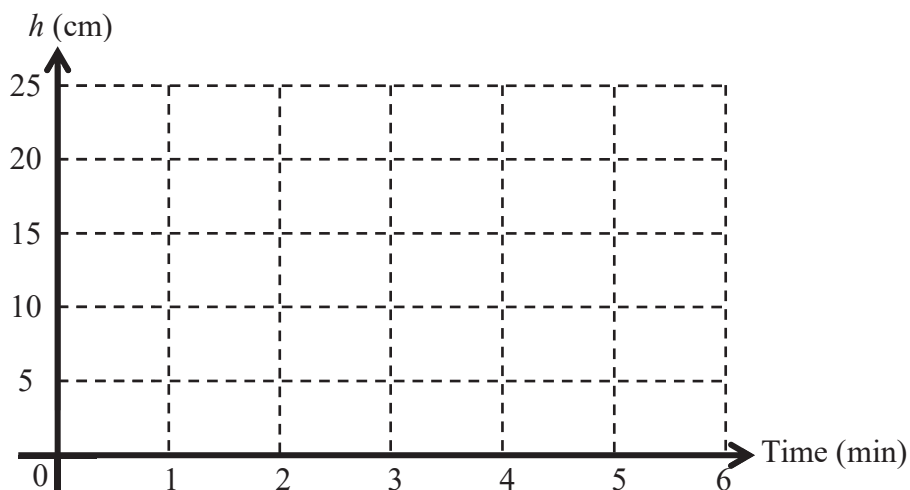
- 9 The container, shown in the diagram, is initially full of water. There is a small hole at the bottom and water is leaking through the hole at a constant rate.



It takes one minute for the water level to drop from Point *A* to Point *B* and another three minutes to drop from Point *B* to Point *C* and a further 90 seconds for the water to be fully drained from the container.

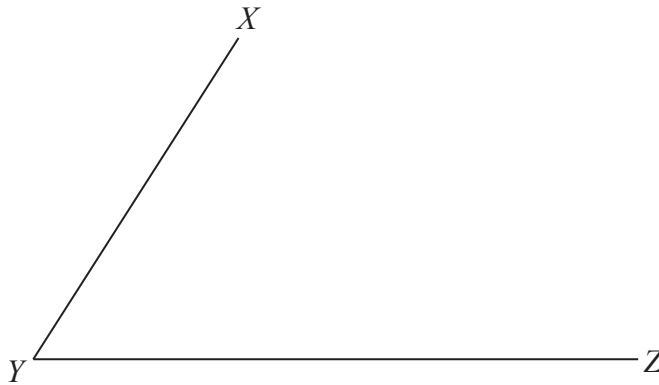
On the axes below, sketch the graph showing the depth of the water, h cm, in the container.

Answer



[3]

- 10 Three points X , Y and Z are shown below.



- (a) P is equidistant from the points Y and Z . [2]
 P is equidistant from the lines YX and YZ .
Construct and label the point P .
- (b) The point W is such that $\angle YXW = 125^\circ$ and WZ is 5 cm. [2]
Find the two possible positions of W and label them W_1 and W_2 .
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